MILITARY SPECIFICATION

# CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the general requirements for button-style (feed-thru and stand-off, hermetically and nonhermetically sealed), mica-dielectric, fixed capacitors for use at frequencies up to 500 megahertz (MHz) in tuning, coupling, and bypass circuits. These capacitors are suitable for operation over a temperature range of -55° to +85°C, or -55° to +150°C, as specified. (See 3.1.)

1.2 Classification.

1.2.1 Part number. The part number shall be in the following form, and as specified (see 3.1 and 6.1):

| CB11 | N | D | 510 | G |
|---|---|---|---|---|
| Style (1.2.1.1) | Terminal assembly (1.2.1.2) | Characteristic (1.2.1.3) | Capacitance (1.2.1.4) | Capacitance tolerance (1.2.1.5) |

1.2.1.1 Style. The style is identified by the two-letter symbol "CB" followed by a two-digit number; the letters identify button-style, mica-dielectric, fixed capacitors and the number identifies the shape, dimensions, and operating-temperature range of the capacitor.

1.2.1.2 Terminal assembly. The terminal assembly is identified by a single letter in accordance with table I.

TABLE I. Terminal assembly.

| Symbol | Terminal assembly |
|---|---|
| L - - - - - - - - | Single L - Rod |
| N - - - - - - - - | Eyelet |
| P - - - - - - - - | Single L |
| Q - - - - - - - - | Single U |
| R - - - - - - - - | Double L |
| S - - - - - - - - | Double U |
| T - - - - - - - - | Offset |
| V - - - - - - - - | Rod |

1.2.1.3 Characteristic. The characteristic is identified by a single letter in accordance with table II.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE II. Characteristic.

| Symbol | Temperature coefficient (PPM/°C) | Capacitance drift (% or pf) |
|---|---|---|
| B | Not specified | Not specified |
| D | -100 to +100 | 0.3 percent or 0.3 pf, whichever is greater |
| E | -20 to +100 | ±(0.1 percent +0.1 pf) |

1.2.1.4 Capacitance. The nominal capacitance value expressed in picofarads (pf) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.5 Capacitance tolerance. The capacitance tolerance in percent is identified by a single letter in accordance with table III.

TABLE III. Capacitance tolerance.

| Symbol | Capacitance tolerance |
|---|---|
| | Percent (±) |
| F | 1 |
| G | 2 |
| J | 5 |
| K | 10 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-39028 - Capacitors, Packaging of, General Specification for.
MIL-C-45662 - Calibration Systems Requirements.

See supplement 1 for list of associated specification sheets or detail specifications.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-174 - Colors for Coding Electronic Parts.
MIL-STD-202 - Test Methods for Electronic and Electric Componen. Parts.
MIL-STD-456 - Electronic Parts, Date and Source Coding for.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern. (See 6.1.)

3.2 Qualification. Capacitors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.2).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Certification that such material has met applicable requirements shall be forwarded by the supplier through the cognizant government inspection element for approval by the procurement activity. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Case and terminals. The capacitor case and terminals shall be of silver-plated brass. Ferrous material may be used as required for styles where specified (see 3.1) if suitably protected with a silver or gold plating or a solder coating.

3.3.2 Soldering flux. Rosin or rosin and alcohol shall be used as a flux. Consideration will be given to the use of other soldering fluxes when there is certification that such other fluxes have met applicable requirements and shall be forwarded, by the supplier through the cognizant government inspection element, for approval by the procurement activity.

3.3.3 Sealing compounds and impregnants. Sealing compounds and impregnants, including varnishes, waxes, and the like, shall be suitable for each particular application. Compounds shall preserve the electrical characteristics of the material to which they are applied, and afford adequate protection against the effects of prolonged exposure to high humidities.

3.4 Design and construction. The capacitors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Threaded parts. All threaded parts shall be class 2A or 2B fit after plating, in accordance with Handbook H28.

3.5 Dielectric withstanding voltage. Capacitors shall withstand the direct-current (dc) potential specified in 4.6.2 without breakdown, flashover, or physical damage.

3.6 Barometric pressure. Capacitors shall withstand the dc potential specified in 4.6.3 without flashover.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

3.7 Insulation resistance. When measured as specified in 4.6.4, the insulation resistance shall be as specified (see 3.1).

3.8 Capacitance. When measured as specified in 4.6.5, the capacitance shall be within the tolerance specified (see 3.1).

3.9 Dissipation factor (DF). When determined as specified in 4.6.6, the DF of the capacitors shall be equal to or less than the value shown grapically on figure 1.

3.10 Shock (medium impact). When capacitors are tested as specified in 4.6.7, there shall be no intermittent contacts of 0.5 milliseconds (ms) or greater duration, or arcing or other indication of breakdown, nor shall there be any open- or short-circuiting or evidence of mechanical damage.

3.11 Vibration. When capacitors are tested as specified in 4.6.8, there shall be no physical damage, intermittent contacts of 0.5 ms or greater duration, or open- or short-circuiting.

3.12 Temperature cycling and immersion. When tested as specified in 4.6.9, capacitors shall meet the following requirements:

| | |
|---|---|
| Dielectric withstanding voltage | As specified in 3.5. |
| Insulation resistance | Not less than the value specified (see 3.1). |
| Capacitance | Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.6.5. |
| Dissipation factor | Equal to or less than 133 percent of value specified in 3.9. |

3.13 Moisture resistance. When tested as specified in 4.6.10, the capacitors shall meet the following requirements:

| | |
|---|---|
| Dielectric withstanding voltage | As specified in 3.5. |
| Insulation resistance | Not less than the value specified (see 3.1). |
| Capacitance | Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.6.5. |
| Dissipation factor | Equal to or less than the percent specified (see 3.1) of the value required in 3.9. |

3.14 Life. When tested as specified in 4.6.11, capacitors shall meet the following requirements:

| | |
|---|---|
| Dielectric withstanding voltage | As specified in 3.5. |
| Insulation resistance | As specified (see 3.1). |
| Capacitance | Change not more than value specified (see 3.1) from the initial value obtained when measured as specified in 4.6.5. |
| Dissipation factor | As specified in 3.9. |

In addition, there shall be no dripping or flowing of coating, or leakage of any kind.

3.15 Temperature coefficient and capacitance drift. The temperature coefficient and capacitance drift shall be as specified in table II for each characteristic (see 4.6.12).

3.16 Torque. When capacitors are tested as specified in 4.6.13, there shall be no perceptible movement of the terminal relative to the case of the capacitor.

3.17 Marking (see 3.1). Capacitors shall be permanently and legibly marked by typographic marking or color coding. Paper labels shall not be used. Other markings which in any way interfere with, obscure, or confuse those specified herein are prohibited. The marking shall remain legible after all tests.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

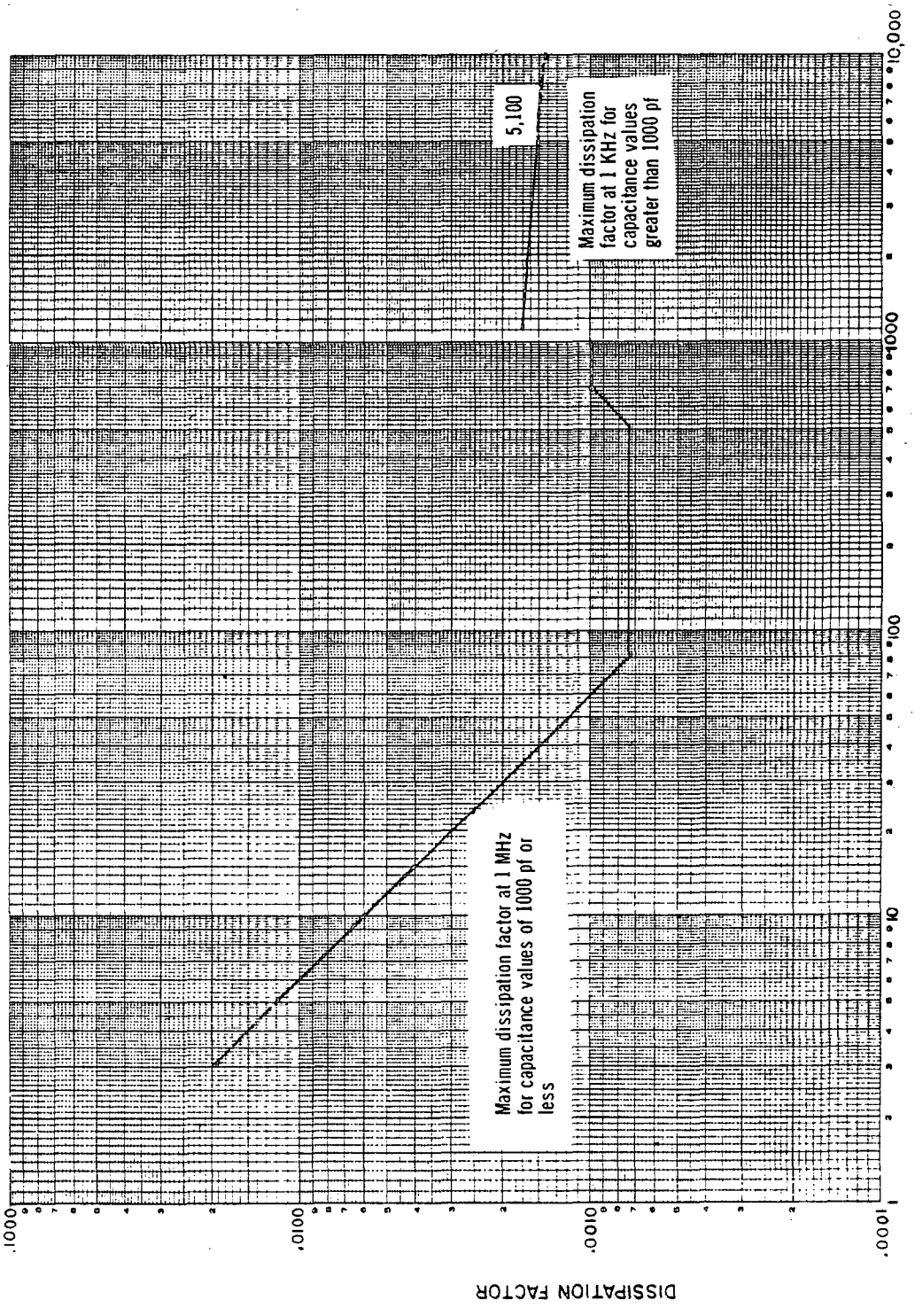



FIGURE 1. Dissipation factor vs capacitance.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

3.17.1 Full marking. Capacitors shall be marked with part number, capacitance (in pf), capacitance tolerance, rated voltage, and the source code. The source code shall be in accordance with MIL-STD-456.

3.17.2 Minimum marking. Where space does not permit marking specified in 3.17.1, capacitors may be marked with part number and source code.

3.17.3 Color coding. Where space does not permit typographic marking, the color code specified in table IV shall be applied (see 3.1). The colors shall be shown in the form of six dots arranged clockwise in the order in which they are to be read (see 3.1). The first dot (MIL identifier) shall be black. Colors shall be in accordance with MIL-STD-174 and shall be electrically nonconductive.

TABLE IV. Color code.

| Color | Capacitance (pf) | | Capacitance tolerance | Characteristic |
|---|---|---|---|---|
| | First and second significant figures | Multiplier 1/ | | |
| | | | Percent (±) | |
| Black | 0 | 1 | --- | --- |
| Brown | 1 | 10 | 1 (F) | B |
| Red | 2 | 100 | 2 (G) | --- |
| Orange | 3 | 1,000 | --- | D |
| Yellow | 4 | --- | --- | E |
| Green | 5 | --- | --- | --- |
| Blue | 6 | --- | --- | --- |
| Purple (violet) | 7 | --- | --- | --- |
| Gray | 8 | --- | --- | --- |
| White | 9 | --- | --- | --- |
| Gold | -- | --- | 5 (J) | --- |
| Silver | -- | --- | 10 (K) | --- |

1/ The multiplier is the factor by which the two significant figures are multiplied to yield the nominal capacitance. (See 1.2.1.4.)

3.18 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.18.1 Riveting. Riveting operations shall be performed carefully to insure that rivets are tight and satisfactorily headed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Qualification inspection. 1/ Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in table V or in the appendix to this specification.

4.4.2 Inspection routine.

4.4.2.1 Single-type and combined-style submission. The sample shall be subjected to the qualification inspection specified in table V, in the order shown. Two sample units shall be subjected to visual and mechanical examination (internal). The remaining sample units shall be subjected to the subsequent inspection of group I. The sample shall then be divided as specified in table V for groups II to V inclusive and subjected to the inspection for their particular group; for combined-style submissions, each style shall be equally represented in each group.

4.4.2.2 Additional-terminal-assembly submission. The sample shall be subjected to the qualification inspection specified in group I of table V, except for visual and mechanical examination (internal), in the order shown. Twelve of the sample units shall then be divided equally into two groups of six units each; one group shall be subjected only to the moisture-resistance test and the other group only to the torque test.

4.4.3 Failures. Failures in excess of those allowed in table V shall be cause for refusal to grant qualification approval.

4.4.4 Retention of qualification. To retain qualification, the supplier shall forward, at 6-month intervals, to the qualifying activity, a summary of the results of groups A and B tests, indicating as a minimum the number of lots which passed and the number which failed, and a summary of the results of group C tests, including the number and type of any part failures. The summary shall include those tests performed during that 6-month period. If the summary of the test results indicates nonconformance with specification requirements, action shall be taken to remove the failing product from the qualified products list. Failure to submit the summary shall result in loss of qualification for that product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification. In the event that no production occurred during the reporting period, a negative report shall be submitted.

1/ Application for qualification tests shall be made in accordance with "Provisions Governing Qualification".



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE V. Qualification inspection.

| Examination or test | Requirement paragraph | Method paragraph | Number of sample units to be inspected | Number of failures allowed | Number of sample units to be inspected for each additional terminal assembly | Number of failures allowed |
|---|---|---|---|---|---|---|
| Group I | | | | | | |
| Visual and mechanical examination (internal): | --- | 4.6.1 | 2 | 1 | 13 [2/] | 1 |
| Material, design, construction, and workmanship | 3.3 to 3.4.1, incl, 3.18, and 3.18.1 | --- | | | | |
| Visual and mechanical examination (external) [4/]: | --- | 4.6.1 | | | | |
| Physical dimensions, marking, [1/], and workmanship | 3.4, and 3.17 to 3.18.1, incl. | --- | | | | |
| Dielectric withstanding voltage [4/] | 3.5 | 4.6.2 | 31 [2/] | | | |
| Barometric pressure [4/] | 3.6 | 4.6.3 | | | | |
| Insulation resistance [4/] | 3.7 | 4.6.4 | | | | |
| Capacitance [4/] | 3.8 | 4.6.5 | | | | |
| DF [4/] | 3.9 | 4.6.6 | | | | |
| Group II [3/] | | | | | | |
| Shock (medium impact) | 3.10 | 4.6.7 | 6 | 1 | | |
| Vibration | 3.11 | 4.6.8 | | | | |
| Temperature cycling and immersion | 3.12 | 4.6.9 | | | | |
| Group III [3/] | | | | | | |
| Moisture resistance | 3.13 | 4.6.10 | 6 | 1 | 6 | 1 |
| Group IV [3/] | | | | | | |
| Life | 3.14 | 4.6.11 | 12 | 1 | | |
| Group V [3/] | | | | | | |
| Temperature coefficient and capacitance drift [4/] | 3.15 | 4.6.12 | 6 | 1 | 6 | 1 |
| Torque | 3.16 | 4.6.13 | | | | |

[1/] Marking will be considered as a defect if illegible initially or as a result of any of the inspection.

[2/] One additional sample unit is included in each sample to permit substitution for the allowable defective in group I.

[3/] In groups II, III, IV, and V, when more than one measurement is made at the end of an inspection or when an inspection consists of two parts, a sample unit having one or more defects will be considered as a single defective.

[4/] Nondestructive examinations and tests.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.5.1.4.1.4, delivery of products which have passed the groups A and B inspections shall not be delayed pending the results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all capacitors of the same group as indicated in table VI produced under essentially the same conditions, and offered for inspection at one time.

TABLE VI. Inspection lot groups.

| Group | Styles |
|---|---|
| 1 | CB11 |
| 2 | CB21, CB31 |
| 3 | CB13, CB14 |
| 4 | CB50 |
| 5 | CB55, CB56, CB57 |
| 6 | CB65, CB66, CB67 |
| 7 | CB60, CB61, CB62 |

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table VII, and shall be made on the same set of sample units in the order shown.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table VII. Major and minor defects shall be defined in MIL-STD-105 and as specified in table VII.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE VII. Group A inspection.

| Examination or test | Requirement paragraph | Method paragraph | AQL (percent defective) | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual and mechanical examination: | --- | 4.6.1 | 1.0 | 4.0 |
| Material | 3.3 to 3.2.3, incl | --- | | |
| Body dimensions | 3.4 | --- | | |
| Design and construction (other than body dimensions) | 3.4 and 3.4.1 | --- | | |
| Marking 1/ | 3.17 and 3.17.1 | --- | | |
| Workmanship | 3.18 and 3.18.1 | --- | | |
| Dielectric withstanding voltage | 3.5 | 4.6.2 | 0.65 | --- |
| Insulation resistance (at room ambient temp only) | 3.7 | 4.6.4 | | |
| Capacitance | 3.8 | 4.6.5 | | |
| DF | 3.9 | 4.6.6 | | |

1/ Marking defects are based on visual examination only. Any subsequent electrical defects shall not be used as a basis for determining marking defects.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.5.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table VIII, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE VIII. Group B inspection.

| Examination or test | Requirement paragraph | Method paragraph |
|---|---|---|
| Barometric pressure | 3.6 | 4.6.3 |
| Insulation resistance (at high ambient temperature) | 3.7 | 4.6.4 |
| Shock (medium impact) | 3.10 | 4.6.7 |
| Vibration | 3.11 | 4.6.8 |
| Temperature cycling and immersion | 3.12 | 4.6.9 |

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. (The sample size selected for group B inspection shall be based on the lot size and not on the group A sample size.) The AQL shall be 4 percent defective.

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.5.1.4 Periodic inspection. Periodic inspection shall consist of group C inspection.

4.5.1.4.1 Group C inspection. Group C inspection shall consist of the tests specified in table IX, in the order shown for each subgroup. Group C inspection shall be made on sample units which have passed groups A and B inspections.

TABLE IX. Group C inspection.

| Examination or test | Requirement paragraph | Method paragraph | Sample units | Number of defectives permitted |
|---|---|---|---|---|
| Subgroup 1 (every month) | | | | |
| Temperature coefficient and capacitance drift | 3.15 | 4.6.12 | 12 | 1 |
| Torque | 3.16 | 4.6.13 | | |
| Subgroup 2 (every month) | | | | |
| Moisture resistance | 3.13 | 4.6.10 | 12 | 1 |
| Subgroup 3 (every 3 months) | | | | |
| Life | 3.14 | 4.6.11 | 12 | 1 |

4.5.1.4.1.1 Sampling plan. Once each calendar month for subgroups 1 and 2 and every 3 months for subgroup 3, sample units shall be taken at random from capacitors currently in production, irrespective of style, terminal assembly, characteristic, capacitance and capacitance tolerance. The number of sample units to be tested for each subgroup shall be as specified in table IX. Different sample units shall be used for each subgroup.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.5.1.4.1.2 Defectives. If the number of defectives exceeds the number allowed in table IX, the sample shall be considered to have failed.

4.5.1.4.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.1.4.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units ( all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.2 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of MIL-C-39028.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3 to 3.4.1, inclusive, and 3.17 to 3.18.1, inclusive).

4.6.2 Dielectric withstanding voltage (see 3.5). Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details and exception shall apply:

(a) Magnitude and nature of test voltage - 200 percent of rated dc working voltage.
(b) Points of application and duration of test voltage - Applied between the terminals of each capacitor for not less than 1 but not more than 5 seconds.
(c) Surge current - Shall not exceed 5 milliamperes (ma).

4.6.3 Barometric pressure (reduced) (see 3.6). Capacitors shall be tested in accordance with method 105 of MIL-STD-202. The following details and exception shall apply:

(a) Method of mounting - Normal mounting means, except that soldering of tabs is not required.
(b) Test condition letter - C.
(c) Tests during subjection to reduced pressure - 125 percent of the dc working voltage (see 3.1) shall be applied between the terminals of each capacitor for not less than 1 but not more than 5 seconds.
(d) Surge current - Shall not exceed 5 ma.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.6.4 Insulation resistance (see 3.7). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter - A (100 volts) or B (500 volts).
(b) Points of measurement - Between the mutually insulated points.
(c) Temperature - Applicable high rated temperature and room ambient temperature.

4.6.5 Capacitance (see 3.8). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details shall apply:

(a) Test frequency -
1 MHz ±100 kilohertz (kHz) - nominal capacitance value 1,000 pf or less.
1 kHz ±100 Hertz (Hz) - nominal capacitance value more than 1000 pf.
(b) Limit of accuracy ±(0.2 percent +0.25 pf).

4.6.6 Dissipation factor (see 3.9). Dissipation factor shall be tested in accordance with method 306 of MIL-STD-202. The following detail shall apply:

(a) Test frequency - As specified in 4.6.5.

4.6.7 Shock, (medium impact) (see 3.10). Capacitors shall be tested in accordance with method 205 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - Normal mounting means.
(b) Number and direction of applied shocks - Twenty (ten in each of two mutually perpendicular directions) (five blows in each of two directions). One of the test planes shall be parallel and the other perpendicular to the cylindrical axis of the capacitor.
(c) Test condition letter - C (50G's).
(d) Electrical load during shock - Rated voltage shall be applied.
(e) Examination during and after shock - During the test, observation shall be made for momentary or intermittent arcing of 0.5 ms or greater duration. After the test, capacitors shall be visually examined for evidence of breakdown, arcing, fractures, loosening of parts, or other visible mechanical damage.

4.6.7.1 Shock (specified pulse). As an option, capacitors may be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - Normal mounting means.
(b) Test condition letter - A (50G's).
(c) Electrical load during shock - Rated voltage shall be applied.
(d) Examination during and after shock - During the test, observation shall be made for momentary or intermittent arcing of 0.5 ms or greater duration. After the test, capacitors shall be visually examined for evidence of breakdown, arcing, fractures, loosening of parts, or other visible mechanical damage.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.6.8 Vibration (see 3.11). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - Normal mounting means.
(b) Test condition - B.
(c) Initial measurements - Not applicable.
(d) Motion - Shall be applied in a direction perpendicular to the cylindrical axis.
(e) Final measurements - During the last 30 minutes of vibration, an electrical measurement shall be made to determine intermittent contacts of 0.5 ms or greater duration or open- or short-circuiting. After this test, capacitors shall be visually examined for physical damage.

4.6.9 Temperature cycling and immersion.

4.6.9.1 Temperature cycling. Capacitors shall be tested in accordance with method 102 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition letter - D, except that step 3 shall be the applicable maximum rated temperature (see 3.1).
(b) Measurements before and after cycling - Not applicable.

4.6.9.2 Immersion. Following the temperature cycling, but not necessarily immediately thereafter, capacitors shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Measurements after final cycling - Within 2 to 6 hours after removal of capacitors from the immersion tank, dielectric withstanding voltage, insulation resistance at room ambient temperature, capacitance, and DF shall be measured as specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6, respectively. If the capacitors fail the insulation resistance test, they may be retested for insulation resistance after a total maximum drying period of 24 hours at a relative humidity of 50±5 percent.

4.6.10 Moisture resistance (see 3.13). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - Except during measurement, normal mounting means.
(b) Initial measurements - Not applicable.
(c) Loading voltage - Not applicable.
(d) Final measurements - Upon completion of step 6 of the final cycle and within 2 to 6 hours after removal of capacitors from the humidity chamber, dielectric withstanding voltage, insulation resistance at room ambient temperature, capacitance, and DF shall be measured as specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6, respectively. If the capacitors fail the insulation-resistance test, they may be retested for insulation resistance after a total maximum drying period of 24 hours at a relative humidity of 50±5 percent.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

4.6.11 Life (at elevated ambient temperature) (see 3.14). Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

(a) Distance of temperature measurements from specimens, in inches - N/A.
(b) Test temperature and tolerance - At the applicable high test temperature, $^{+3^\circ}_{-0^\circ}$ C. (See 3.1.)
(c) Operating conditions - A dc potential of 150 percent of the rated dc working voltage shall be applied continuously. The surge current shall not exceed 5 ma. The maximum alternating current ripple current superimposed on the dc voltage shall not exceed 5 percent.
(d) Test-condition letter - F.
(e) Measurements - After the application of voltage, the capacitors shall be returned to the inspection conditions specified in 4.3, and dielectric withstanding voltage, insulation resistance at room ambient temperature, capacitance and DF shall then be measured as specified in 4.6.2, 4.6.4, 4.6.5, and 4.6.6 respectively.

4.6.12 Temperature coefficient and capacitance drift (see 3.15).

4.6.12.1 For qualification inspection. Capacitance measurements shall be made at the following temperatures, in the order listed.

| For capacitors whose operating temperature range is -55 to +85° C<br>° C $^{+2^\circ}_{-0^\circ}$ | For capacitors whose operating temperature range is -55 to +150° C<br>° C $^{+2^\circ}_{-0^\circ}$ |
|---|---|
| 25 | 25 |
| -55 | -55 |
| -10 | -40 |
| 1/ 25 | 1/ 25 |
| 45 | 45 |
| 65 | 65 |
| 85 | 85 |
| 25 | 105 |
| | 125 |
| | 150 |
| | 25 |

1/ This 25° C temperature shall be considered as the reference temperature.

The measurement at each temperature shall be recorded when two successive readings taken at 5-minute intervals at that temperature indicate no change in capacitance. Measurements shall be made at a frequency as specified in 4.6.5. The resolution of measurement of ±(0.025 percent of nominal capacitance +0.05 pf) shall be maintained for measurement of capacitance increments.

4.6.12.1.1 Temperature coefficient. The temperature coefficient shall be computed as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

$$TC = \frac{(C_2 - C_1)\ 10^6}{(T_2 - T_1)\ C_1}$$

Where:

TC = temperature coefficient in parts per million per degree C.
$C_1$ = capacitance at 25° C in pf.
$C_2$ = capacitance at test temperature in pf.
$T_1$ = 25° C.
$T_2$ = test temperature in degrees C.

4.6.12.1.2 Capacitance drift. Capacitance drift in percent shall be computed by dividing the greatest single difference between any two of the three values recorded at 25°C by the second value recorded at 25° C, multiply this ratio by 100. Capacitance drift in pf shall be computed by subtracting the smallest capacitance value recorded from the largest of the three values recorded at 25° C.

4.6.12.2 For quality conformance inspection.

4.6.12.2.1 Temperature coefficient. Capacitance shall be measured as specified in 4.6.12.1, except that measurements shall be made only at the following temperatures, in the order listed:

| For capacitors whose operating temperature range is -55° to +85° C<br>° C $^{+2°}_{-0°}$ | For capacitors whose operating temperature range is -55° to +150° C<br>° C $^{+2°}_{-0°}$ |
|---|---|
| 25 | 25 |
| -55 | -55 |
| 1/ 25 | 1/ 25 |
| 85 | 150 |
| 25 | 25 |

1/ This 25° C temperature shall be considered as the reference temperature.

The temperature coefficient shall be computed in accordance with 4.6.12.1.1.

4.6.12.2.2 Capacitance drift. Capacitance drift shall be computed in accordance with 4.6.12.1.2.

4.6.13 Torque. Capacitors shall be mounted as specified (see 3.1). In addition, capacitors with N-type terminal assemblies shall have an AWG size 18 solid-copper wire inserted through the eyelet and soldered into position so that the wire extends at least 1 inch perpendicular to one face of the capacitor. A torque of 2 ounce-inches for N-type terminal assemblies and of 4 ounce-inches for all other types of terminal assemblies shall then be applied to the terminals. The torque shall be applied so that it will tend to produce clockwise or counterclockwise rotation of the terminal or wire relative to the case about the axis of symmetry of the capacitor. The torque shall be applied in such a manner as to minimize bending of the terminal. (See 3.16.)

## 5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Capacitors shall be prepared for delivery in accordance with MIL-C-39028.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet and the complete part number. (See 1.2.1 and 3.1.)
(c) Required levels of preservation and packaging and packing. (see section 5.)

6.1.1 Indirect shipment. The preservation, packaging, packing, and marking requirements (see section 5) apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Qualification.[2/] With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, Directorate of Engineering Standardization (DESC-E) Dayton, Ohio 45401.

6.3 Mounting. It is recommended that the capacitor body have adequate heat sink during mounting operation with high temperature solder.

6.4 Cross-reference of styles. The styles covered by this specification are mutually interchangeable for styles covered by the superseded specification. Dash number to part number conversion data is specified in the applicable specification sheet.

6.5 Selection and use information. Equipment designers should refer to MIL-STD-198[2/] for a selection of standard capacitor types and values for new equipment design. Application and use information concerning these capacitors is also provided in MIL-STD-198.

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement (NEPR No. 12). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

[2/] Copies of "Provisions Governing Qualification" and MIL-STD-198, "Capacitors, Selection and Use of," may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Preparing activity:
Navy - EC

(Project 5910-0847)

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Navy - AS, OS, CG



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

# APPENDIX

## QUALIFICATION-SUBMITTAL PLANS

### 10. SCOPE

10.1 This appendix details the qualification-submittal plans for capacitors to be subjected to the qualification inspection specified in this specification. These plans specify the sample size, sampling criteria, and limits of coverage for the type of qualification sought.

### 20. SUBMITTAL PLANS

20.1 Single type. The following details apply:

(a) Sample size - As specified in table V.
(b) Sampling criteria - Sample units of the same style, terminal assembly, characteristic, capacitance value, and capacitance tolerance.
(c) Limits of coverage - Based upon qualified sample units, qualification coverage will be extended as follows:
  (1) Style - Restricted to style qualified.
  (2) Terminal assembly - Restricted to terminal assembly qualified; however, additional coverage may be gained in accordance with table V.
  (3) Characteristic - All characteristics equal to and wider than the characteristic qualified.
  (4) Capacitance value - All capacitance values equal to and less than the value qualified.
  (5) Capacitance tolerance - All capacitance tolerances equal to and wider than the tolerance qualified.

20.2 Combined-style type. The following details apply:

(a) Sample size - As specified in table XI.
(b) Sampling criteria - Sample units of (1) basically the same construction (seal, body and terminal assembly configuration) and (2) sample units of the highest capacitance value, tighest tolerance and characteristic within a style.
(c) Limits of coverage - As specified in 20.1(c), except that coverage will be extended to all intermediate styles and all terminal assemblies.

NOTE: Single-type submittal plan shall be used for style CB50.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

TABLE XI. Combined-style submittal data.

| Styles | Type designation | Sample size |
|---|---|---|
| CB11 and CB14 | CB11NE152F<br>CB14SE512F | 18<br>18 |
| CB21 and CB31 | CB21TE242F<br>CB31QE242F | 18<br>18 |
| CB60 and CB62 | CB60RE471F<br>CB62RE242F | 18<br>18 |
| CB55 and CB57 | CB55RE471F<br>CB57RE242F | 18<br>18 |
| CB65 and CB67 | CB65PE471F<br>CB67PE242F | 18<br>18 |

30. DATA REQUIREMENTS

30.1 Description of items. The supplier shall submit a detailed description of the capacitors being subjected to qualification inspection, including material, thickness and finish of case, type and thickness of dielectric, type of electrodes, type of impregnant, and details of terminal assemblies.

30.2 Certification of material. The supplier shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

30.3 Test data (not required for in-plant testing). Prior to submission, all sample units shall be subjected to all of the nondestructive examinations and tests (see table V). Each submission shall be accompanied by the test data obtained from these examinations and tests. The performance of the destructive examinations and tests by the supplier on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |
| | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:38Z
Check the source to verify that this is the current version before use.